

SOLUTION REQUIREMENTS

OptiCrop: Smart Agricultural Production Optimization Engine

Date	03 July 2026
Team ID	SWTID-2026-7253
Team Size	5
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Maximum Marks	4 Marks

Introduction

The Solution Requirements define the functional capabilities and expected behavior of the **OptiCrop: Smart Agricultural Production Optimization Engine**. These requirements specify the services provided by the application, user interactions, system operations, data processing, and machine learning functionality required to deliver accurate and efficient agricultural optimization recommendations. By integrating environmental factors such as Nitrogen (N), Phosphorous (P), Potassium (K) levels, temperature, humidity, pH, and rainfall, the platform provides farmers, researchers, and policymakers with evidence-based agricultural strategies.

1. Functional Requirements (FR)

FR ID	Requirement	Description	Priority
FR-01	User Registration	Applicants/farmers shall register securely using personal and regional profile information.	High
FR-02	User Login	Registered users shall authenticate using valid credentials.	High
FR-03	Profile Management	Users shall update their personal, regional, and farm profile information.	Medium
FR-04	Environmental Parameter Submission	Farmers shall submit soil metrics (N, P, K, pH) and climate values (temperature, humidity, rainfall).	High
FR-05	Input Validation	The system shall validate mandatory environmental fields before execution and submission.	High
FR-06	Data Preprocessing	The system shall clean and transform incoming parameter matrices before recommendation modeling.	High
FR-07	Optimization Engine Prediction	The system shall recommend the most suitable crop or predict compatibility scores using the trained ML model.	High
FR-08	Prediction History Log	Store previous optimization assessments and crop recommendation logs.	Medium

FR ID	Requirement	Description	Priority
FR-09	Administrator & Researcher Dashboard	Administrators and researchers shall monitor regional data distribution and optimization metrics.	Medium
FR-10	Report Generation & Export	Generate downloadable crop compatibility and regional production analytics reports.	Medium

Functional Requirements Summary: The functional requirements define the essential services required for secure user management, environmental parameter submission, intelligent prediction, reporting, and system administration.

2. Non-Functional Requirements (NFR)

NFR ID	Requirement	Description	Acceptance Criteria
NFR-01	Performance	Crop recommendations and compatibility metrics shall be generated quickly.	Response time less than 2 seconds.
NFR-02	Model Accuracy	The predictive engine shall provide highly reliable crop optimization matches.	Prediction accuracy greater than 90% (ROC AUC benchmark).
NFR-03	Security	Protect regional user and agricultural metrics using secure authentication and data encryption.	HTTPS, encrypted data payloads, role-based access control.
NFR-04	Availability	The software suite shall remain continuously available during seasonal operations.	Minimum 99.9% application uptime.
NFR-05	Scalability	Effectively support an increasing number of registered farmers and institutional researchers.	Support scalable concurrent active connections.
NFR-06	Usability	Provide a simple, responsive, and intuitive graphical user interface.	Easy navigation structure and responsive design formats.
NFR-07	Reliability	Ensure ongoing stable execution of the agricultural analytics core.	Minimal prediction system failures.
NFR-08	Maintainability	Support future optimization enhancements, geographical rules, and model retraining cycles.	Modular component design structure.

3. System Assumptions

- Users provide accurate soil data metrics and local climate parameters.
- Active internet or regional network connectivity is available for remote model calls.
- The pre-trained agricultural production machine learning model is successfully initialized and deployed.
- The relational database server operates normally without query lockout constraints.

- Authorized planning and analytical panels are strictly restricted to verified administrative/research accounts.

4. System Constraints

- Recommendation quality remains bound by the statistical depth and quality of the training datasets.
- A working internet connection is required to communicate with central cloud intelligence nodes.
- Highly incomplete or unsupported out-of-bounds metrics may negatively affect prediction confidence intervals.
- Core software execution speed depends directly on allocated hardware server node resources.
- Future system model adaptation requires periodic data refresh cycles reflecting evolving seasonal variations.

Conclusion

The Solution Requirements establish the functional and non-functional expectations of the **OptiCrop: Smart Agricultural Production Optimization Engine**. These requirements ensure the application remains secure, scalable, reliable, user-friendly, and highly capable of delivering precise crop optimization intelligence using advanced data analytics and machine learning techniques.